AI-Powered Live Session Analysis

Real-time engagement analysis and coaching for video tutoring sessions

Background

Live tutoring sessions are the core value proposition of the platform, yet tutors often lack real-time feedback on their teaching effectiveness. Engagement metrics like eye contact, speaking time balance, and interaction patterns are strong predictors of session quality but are invisible during the session itself.

Your challenge: Create a live, AI-powered system that analyzes active video calls to measure engagement metrics and provides real-time suggestions or flags to help tutors improve session quality.

Project Overview

Individual or team project focused on real-time video analysis, computer vision, and coaching systems.

Deliverables

- [ ] Real-time video stream analysis pipeline
- [ ] Engagement metric calculation (eye contact, speaking time, etc.)
- [ ] Non-intrusive tutor notification system
- [ ] Post-session analytics dashboard
- [ ] Tutor coaching recommendations engine

Core Objectives

- [ ] Analyze video streams in real-time with minimal latency
- [ ] Calculate meaningful engagement metrics accurately
- [ ] Deliver actionable suggestions without disrupting the session
- [ ] Help tutors improve their teaching effectiveness over time

Users

- **Primary:** Tutors conducting live sessions
- **Secondary:** Students (indirect beneficiaries), Quality assurance team

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Core Requirements

1. Real-Time Video Analysis

Process video streams with low latency for live feedback.

Specifications:

- Face detection and tracking for tutor and student
- Eye gaze estimation and attention detection
- Latency under 500ms for real-time feedback
- Handle variable video quality gracefully

2. Engagement Metrics

Calculate actionable engagement indicators.

Specifications:

- **Eye Contact:** Percentage of time participants look at camera/screen
- **Speaking Time:** Balance between tutor and student talk time
- **Interruptions:** Detection and counting of speaking overlaps
- **Energy Level:** Voice tone and facial expression analysis
- **Attention Drift:** Detection of distraction or disengagement

3. Real-Time Coaching

Provide non-intrusive suggestions during sessions.

Specifications:

- Subtle visual indicators (not disruptive to session flow)
- Contextually appropriate timing for suggestions
- Configurable sensitivity and notification frequency
- Examples: "Student hasn't spoken in 5 minutes", "Try making more eye contact"

4. Post-Session Analytics

Comprehensive session review and improvement tracking.

Specifications:

- Session summary with key metrics
- Trend analysis across multiple sessions
- Specific moments flagged for review
- Personalized improvement recommendations

Inputs & Outputs

Inputs

- **Video Streams:** Live video feeds from tutor and student
- **Audio Streams:** Session audio for speaking analysis
- **Session Context:** Subject, duration, student level

Outputs

- **Live Metrics:** Real-time engagement scores
- **Coaching Nudges:** Contextual suggestions during session
- **Session Report:** Post-session analytics summary
- **Improvement Plan:** Personalized tutor development recommendations

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Technical Architecture

Modular Structure

- `video-processor/` - Real-time video analysis pipeline
- `metrics-engine/` - Engagement metric calculations
- `coaching-system/` - Real-time suggestion generation
- `analytics-dashboard/` - Post-session reporting
- `docs/` - Decision log and API documentation

Performance Requirements

- Video processing latency: <500ms
- Metric update frequency: 1-2 Hz
- System resource usage: Reasonable for typical hardware

Success Criteria

Category	Metric	Target
Performance	Analysis latency	<500ms
Accuracy	Eye contact detection accuracy	85%+
Accuracy	Speaking time measurement	95%+
UX	Tutor satisfaction with coaching	4/5+ rating
Impact	Session quality improvement	Measurable increase
Reliability	System uptime during sessions	99.5%+

Ambiguous Elements (You Must Decide)

- How intrusive should real-time coaching be? (Subtle vs. explicit)
- What metrics matter most for different session types?
- How to handle poor video quality or connectivity issues?
- Privacy considerations for video analysis

Technical Contact

For questions or clarifications:

[TBD] - [TBD]

Evaluation Criteria: AI-Powered Live Session Analysis

This document outlines how submissions will be evaluated.

Assessment Overview

Submissions are evaluated across five areas, with emphasis on real-time performance and practical coaching value.

Area	Weight	Focus
Real-Time Performance	25%	Latency, reliability, resource efficiency
Metric Accuracy	25%	Precision of engagement measurements
Coaching Value	20%	Usefulness and timing of nudges/feedback
Technical Implementation	15%	Architecture, code quality, scalability
Documentation	15%	Decision log, limitations, privacy analysis

1. Real-Time Performance (25%)

Excellent (23-25 points)

- Analysis latency consistently <300ms
- Smooth metric updates (1-2 Hz minimum)
- Handles video quality variations gracefully
- CPU usage reasonable for typical hardware
- No dropped frames or audio gaps

Good (18-22 points)

- Analysis latency <500ms

- Metric updates at 1 Hz
- Handles most video conditions
- Acceptable resource usage
- Minimal dropped data

Acceptable (13-17 points)

- Analysis latency <1 second
- Metric updates every 2-3 seconds
- Some issues with poor video
- Higher resource usage
- Occasional dropped data

Needs Improvement (0-12 points)

- Latency >1 second or highly variable
- Infrequent metric updates
- Fails with video quality issues
- Excessive resource usage
- Unreliable data capture

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2. Metric Accuracy (25%)

Excellent (23-25 points)

- Eye contact detection accuracy ≥85%
- Speaking time measurement ≥95% accurate
- Reliable speaker diarization
- Interruption detection with low false positives
- Metrics validated against ground truth

Good (18-22 points)

- Eye contact detection accuracy ≥75%
- Speaking time measurement ≥90% accurate
- Good speaker diarization
- Reasonable interruption detection
- Some validation performed

Acceptable (13-17 points)

- Eye contact detection accuracy ≥65%

- Speaking time measurement ≥85% accurate
- Basic speaker separation
- Interruption detection works
- Limited validation

Needs Improvement (0-12 points)

- Eye contact detection unreliable
- Speaking time inaccurate
- Speaker diarization fails
- Interruption detection broken
- No validation

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3. Coaching Value (20%)

Excellent (18-20 points)

- Nudges are actionable and specific
- Timing is appropriate (not disruptive)
- Configurable sensitivity levels
- Post-session insights are valuable
- Tutors would find system helpful

Good (14-17 points)

- Nudges provide useful guidance
- Generally good timing
- Some configurability
- Post-session summary useful
- Positive tutor value

Acceptable (10-13 points)

- Basic nudge functionality
- Sometimes mistimed
- Limited configuration
- Basic post-session summary
- Some tutor value

Needs Improvement (0-9 points)

- Nudges not useful or absent

- Disruptive or poorly timed
- No configuration
- No post-session insights
- Would not help tutors

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4. Technical Implementation (15%)

Excellent (14-15 points)

- Clean, modular architecture
- One-command setup
- 15+ tests with good coverage
- Handles edge cases gracefully
- Well-documented API/interfaces

Good (11-13 points)

- Reasonable architecture
- Setup works with minimal friction
- 10+ tests covering core functionality
- Most edge cases handled
- Adequate documentation

Acceptable (8-10 points)

- Basic structure that works
- Setup requires some manual steps
- Minimal tests (5-10)
- Some edge case issues
- Basic documentation

Needs Improvement (0-7 points)

- Disorganized code
- Difficult to set up
- Few or no tests
- Many edge case failures
- Poor documentation

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5. Documentation (15%)

Excellent (14-15 points)

- Comprehensive decision log
- Privacy analysis and recommendations
- Clear limitations documented
- Calibration methodology explained
- Easy for others to extend

Good (11-13 points)

- Decision log covers main choices
- Privacy considered
- Key limitations noted
- Some calibration documented
- Understandable by others

Acceptable (8-10 points)

- Basic documentation
- Limited privacy analysis
- Few limitations mentioned
- Minimal calibration info
- Some gaps

Needs Improvement (0-7 points)

- Missing documentation
- Privacy not addressed
- No limitations acknowledged
- No calibration info
- Hard to understand

Scoring Rubric Summary

Score Range	Grade	Description
90-100	Excellent	Exceptional work, exceeds expectations
80-89	Good	Strong work, meets all core requirements well

Score Range	Grade	Description
70-79	Acceptable	Satisfactory work, meets basic requirements
60-69	Needs Work	Partially complete, missing key elements
<60	Incomplete	Does not meet minimum requirements

Automatic Deductions

- **No working demo:** -10 points
- **Cannot run with provided instructions:** -10 points
- **Latency >2 seconds:** -10 points
- **No real-time component:** -15 points
- **No coaching nudges:** -10 points

Bonus Points (up to 10)

- **Browser-based implementation** (no install required): +3
- **Multi-participant support** (group sessions): +3
- **Exceptional visualization** of real-time metrics: +2
- **Novel engagement metrics** beyond specified: +2

Test Scenarios

Your submission will be evaluated against:

- Normal quality webcam video (720p, 30fps)
- Poor quality video (low light, compression artifacts)
- Variable audio quality
- Sessions with different talk time ratios
- Sessions with engagement changes over time

You should create your own test videos for development.

Submission Checklist

Before submitting, verify:

- [ ] Code runs with one command (or clear, minimal setup)
- [ ] README explains setup and usage
- [ ] Real-time latency measured and reported
- [ ] Metric accuracy validated and documented
- [ ] Coaching nudges functional and configurable
- [ ] Post-session analytics available
- [ ] Privacy considerations documented
- [ ] Decision log documents major choices
- [ ] Limitations are explicitly stated
- [ ] Demo video or live walkthrough included

Starter Kit: AI-Powered Live Session Analysis

This document provides resources, examples, and guidance to help you get started.

Problem Context

Live tutoring sessions are high-value interactions, but tutors often lack real-time feedback on their teaching effectiveness. Key engagement signals like eye contact, talk time balance, and energy levels are invisible during the session itself.

Your system should provide actionable insights without disrupting the teaching flow.

Key Engagement Metrics

1. Eye Contact / Attention

What to measure:

- Percentage of time participant looks at camera/screen
- Gaze direction (looking away, down at notes, at phone)
- Mutual attention (both looking at each other)

Why it matters:

- Strong predictor of engagement and connection
- Indicates whether explanations are landing
- Helps identify distraction or confusion

Challenges:

- Camera angle variations
- Multi-monitor setups
- Looking at shared content vs. disengagement

2. Speaking Time Balance

What to measure:

- Ratio of tutor talk time to student talk time
- Length of speaking turns
- Response latency (time to respond after other speaks)

Why it matters:

- Too much tutor talk = passive learning
- Too little = not enough guidance
- Ideal ratio varies by session type

Benchmarks:

- Lecture/explanation: 70-80% tutor
- Practice/review: 30-50% tutor
- Socratic discussion: 40-60% tutor

3. Interruptions

What to measure:

- Overlapping speech detection
- Who interrupts whom
- Interruption frequency over time

Why it matters:

- High interruption = poor turn-taking
- May indicate confusion or disengagement
- Pattern changes can signal issues

4. Energy Level

What to measure:

- Voice volume and variation
- Facial expression valence
- Speech rate and enthusiasm markers

Why it matters:

- Low energy may indicate boredom or fatigue
- Mismatched energy levels suggest disconnect
- Energy drops may correlate with confusion

5. Attention Drift

What to measure:

- Sudden changes in engagement metrics
- Prolonged lack of response
- Physical indicators (slouching, looking away)

Why it matters:

- Early warning of lost attention
- Opportunity for tutor intervention
- Identifies difficult content areas

Technical Approaches

Video Analysis Pipeline

```
Video Stream → Frame Extraction → Face Detection →
  → Gaze Estimation → Attention Score
  → Expression Analysis → Energy Score
  → Body Pose → Engagement Indicators
```

Audio Analysis Pipeline

```
Audio Stream → Voice Activity Detection →
  → Speaker Diarization → Talk Time Metrics
  → Overlap Detection → Interruption Counts
  → Prosody Analysis → Energy/Enthusiasm Score
```

Real-Time Architecture

```
┌──────────────┐     ┌──────────────┐     ┌──────────────┐
│ Video/Audio  │────►│  Processing  │────►│  Metrics     │
│   Stream     │     │   Pipeline   │     │  Dashboard   │
└──────────────┘     └──────────────┘     └──────────────┘
                           │
                           ▼
                     ┌──────────────┐
                     │   Coaching   │
                     │   Nudges     │
                     └──────────────┘
```

Coaching Nudge Examples

Subtle Indicators (Non-Disruptive)

- Small icon changes in corner of screen
- Color-coded status indicators
- Gentle audio cues (optional)

Nudge Types

Trigger	Nudge Message	Timing
Student silent >3 min	"Check for understanding"	After 3 min silence
Low eye contact	"Student may be distracted"	After 30s pattern
Tutor talk >80%	"Try asking a question"	After 5 min lecture
Energy drop	"Consider a short break"	After 20% drop
Interruption spike	"Give more wait time"	After 3+ interruptions

Nudge Design Principles

1. **Non-intrusive:** Don't break teaching flow
2. **Actionable:** Suggest specific behavior
3. **Timely:** Deliver when correction is possible
4. **Configurable:** Let tutors adjust sensitivity
5. **Private:** Only visible to tutor

Sample Data Structures

Real-Time Metrics

```
{
  "timestamp": "2024-01-15T14:32:45Z",
  "session_id": "session_123",
  "metrics": {
    "tutor": {
      "eye_contact_score": 0.85,
      "talk_time_percent": 0.65,
```

```
      "energy_score": 0.72,
      "current_speaking": true
    },
    "student": {
      "eye_contact_score": 0.45,
      "talk_time_percent": 0.35,
      "energy_score": 0.58,
      "current_speaking": false
    },
    "session": {
      "interruption_count": 2,
      "silence_duration_current": 0,
      "engagement_trend": "declining"
    }
  }
}
```


Coaching Nudge


```
{
  "timestamp": "2024-01-15T14:32:45Z",
  "nudge_type": "engagement_check",
  "message": "Student hasn't spoken in 4 minutes. Consider asking a question.",
  "priority": "medium",
  "trigger_metrics": {
    "student_silence_duration": 240,
    "student_eye_contact_avg": 0.42
  }
}
```


Post-Session Summary


```
{
  "session_id": "session_123",
  "duration_minutes": 45,
  "summary": {
    "talk_time_ratio": {
      "tutor": 0.62,
      "student": 0.38
    },
    "avg_eye_contact": {
      "tutor": 0.78,
      "student": 0.54
    },
```

```
    "total_interruptions": 8,
    "engagement_score": 72,
    "key_moments": [
      {
        "timestamp": "00:12:34",
        "type": "attention_drop",
        "description": "Student engagement dropped significantly"
      }
    ]
  },
  "recommendations": [
    "Try shorter explanation segments",
    "Ask more check-for-understanding questions"
  ]
}
```

Getting Started Checklist

- [ ] Set up video stream capture (webcam or test videos)
- [ ] Implement face detection pipeline
- [ ] Build gaze estimation module
- [ ] Create speaker diarization for audio
- [ ] Calculate basic engagement metrics
- [ ] Design tutor-facing dashboard
- [ ] Implement coaching nudge system
- [ ] Build post-session analytics view
- [ ] Test end-to-end with sample sessions
- [ ] Optimize for real-time performance (<500ms latency)

Resources

Face Detection & Gaze

- MediaPipe Face Mesh
- OpenCV DNN face detection
- GazeML / OpenGaze
- Dlib facial landmarks

Audio Analysis

- pyAudioAnalysis
- Resemblyzer (speaker embeddings)
- WebRTC VAD
- speechbrain

Real-Time Processing

- OpenCV video capture
- WebRTC for browser integration
- FFmpeg for stream handling
- asyncio for concurrent processing

Visualization

- Plotly / Dash for dashboards
- Chart.js for web UIs
- Streamlit for rapid prototyping

Privacy Considerations

- **Consent:** Tutors and students must consent to analysis
- **Data Retention:** Define what's stored and for how long
- **Access Control:** Who can view session analytics?
- **Anonymization:** Consider aggregate vs. individual metrics
- **Transparency:** Clear disclosure of what's being measured